

1(a)	any two from: 1 use coloured filter / set wavelength ; 2 calibrate colorimeter / <u>set</u> colorimeter (absorbance) to zero ; 3 water (used to calibrate colorimeter) ;	2
1(b)(i)	<i>From description or table:</i> 1 Correct method to dilute stock solution with water to 0.1% ; <i>Table:</i> 2 At least five stated concentrations including 0.1%, all with correct volumes of stock and water including volume units and concentration unit ;	2
1(b)(ii)	<i>independent variable</i> <u>protein</u> concentration and <i>dependent variable</i> absorbance ;	1
1(b)(iii)	1 x-axis protein <u>concentration</u> and % and y-axis absorbance and au or no units added ; 2 line drawn showing positive correlation ;	2
1(b)(iv)	copper(II) ions, binding with the nitrogen atoms, that form peptide bonds ; will have same number of peptide bonds ;	2

Question	Answer	Marks
1(c)	any six from: 1 same / stated, volume of protein concentrations and egg ; 2 same / stated, volume of biuret solution ; 3 add / mix / stir, protein (concentrations) / egg and, biuret ; 4 measure / record, absorbance / colorimeter reading, for (known) protein (concentrations) / egg ; 5 use calibration curve to find protein conc. in egg ; 6 dilute egg (suspension) by factor of 10 ; 7 same / stated, concentration of biuret solution ; 8 same / stated, time (for colour development) before taking reading ; 9 3 replicates and finding mean ;	6

Question	Answer		Marks
1(c)	10 named hazard and risk and precaution ;		
	Hazard	Risk	
	Biuret solution	Corrosive	
		Irritant	
		Moderate / medium, hazard / risk	
	Sodium hydroxide	Corrosive	
		Irritant	
	Copper sulphate	Corrosive	
		Irritant	
		Moderate / medium, hazard / risk	
		Hazardous to the aquatic environment	Do not pour down sink
	Egg / Albumin / Protein	Allergen	Wear gloves / goggles / mask / PPE / do not consume
	11 multiply protein concentration in egg, by dilution factor (to obtain actual concentration of protein) ;		